TH1 · DYNAMIC-SDOF

SDOF anchor for THA time integration (Newmark average-acceleration, dt convergence)

Production THA (Newmark) pipeline — §1 SDOF anchor vs independent closed form + §2 MDOF modal-vs-Rayleigh self-consistency

PASS

error −0.0017% · tol 0.01%

§2 TECHNICAL FEATURES TESTED

- The app's own **THA builder** end to end (Newmark 0.5/0.25, piecewise-linear ground motion) on the **STRIX engine**
- §1: independent closed-form SDOF recurrence (undetermined coefficients), cross-checked vs RK4 pre-run
- §1: $O(dt^2)$ convergence order measured directly, not assumed
- §2: SM5 Bathe-Wilson 9-storey frame through the same builder (production node/element/mass emission)
- §2: Rayleigh a0/a1 calibrated from this model's own modal periods (reported by the THA run), not reused

§3 PROBLEM DESCRIPTION

THA (time-history analysis) has no third-party published example in STRIX's scope the way CSI 1-020/1-026 served the RSA/Pushover benchmarks — CSI 1-022's ground-motion record was never digitized (devplan §12.12, explicit user decision), and a full nonlinear-hinge THA is already covered by SH1 (element mechanics) + SP1 (pushover driver). TH1 instead isolates and checks the ONE thing SH1/SP1 do not: the **time-integration scheme itself**, in two complementary stages. **§1 (SDOF anchor)** drives a single isolated single-degree-of-freedom column (one elastic column, base fixed, top free only laterally, lumped top mass) with a resonant sine pulse through the app's production THA builder and the STRIX engine, and compares it — at six halvings of dt — against a closed-form solution re-derived from scratch (undetermined coefficients on the piecewise-linear-interpolated ground motion the builder uses) and cross-validated against an independent RK4 integrator before any engine run (hand-derivation alone is not trustworthy — devplan §12.11/§12.13). The builder's other stages (a gravity stage with no load, tiny masses on restrained DOFs, the recovery ladder) do not act on this problem; every run is checked to advance exactly one output row per dt. **§2 (MDOF self-convergence)** then exercises a REAL multi-mode structure — SM5's already-validated Bathe & Wilson frame — and checks that STRIX's two independent production damping formulations, modal damping (3% on 9 modes) and Rayleigh damping, agree with each other when both are calibrated to the same 3% ratio at the structure's own two dominant frequencies (taken from the periods this THA run itself reports, not reused from SM5's shear-rigid eigenvalues — devplan §12.14). No external reference exists for a full MDOF THA, so §2 is graded as bounded self-consistency (gap must shrink and floor out with dt, target <0.5% at the finest dt), one notch looser than §1's convergence-to-truth proof.

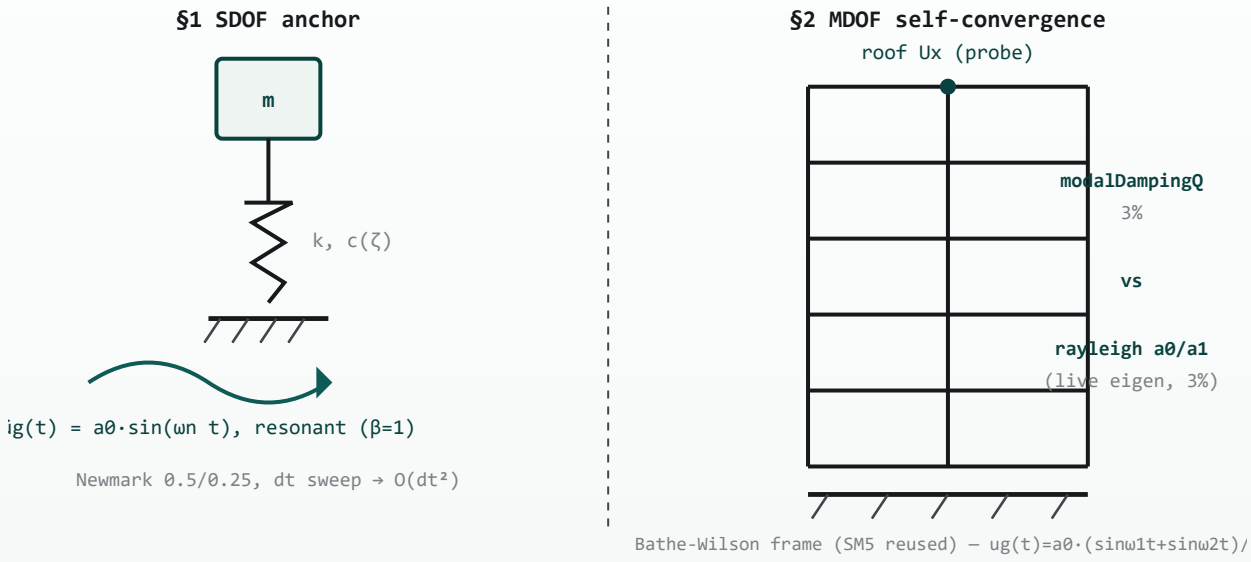


Figure TH1. Left (§1): isolated SDOF spring-mass-damper driven by a resonant ground pulse, dt-swept against an independent closed form. Right (§2): SM5's Bathe-Wilson frame, roof Ux compared between modal and Rayleigh damping calibrated at the frame's own periods.

§4 GEOMETRY, PROPERTIES & LOADING

§1 SDOF RIG

Period T	1 s
Mass m	1 N·s²/mm
Stiffness k	39.4784176044 N/mm
zeta configs	0% (undamped), 5%

§2 MDOF RIG

Model	SM5 Bathe-Wilson (reused)
numModes	9 (production formula)
Damping ratio	3% modal AND Rayleigh
Rayleigh calib.	own modal periods (T1, T2)

ANALYSIS & ENGINE

Integrator	Newmark 0.5/0.25
Ground motion	piecewise-linear, starts at 0
dt sweep (§1 / §2)	6 / 5 halvings
Engine	STRIX engine (THA)

§5 REFERENCE SOLUTION

Peak relative displacement, zeta=5%, dt=0.003125s (finest): max|u(t)|, SDOF recurrence for piecewise-linear-interpolated forcing (undetermined coefficients), zeta=5% = 200.786209 mm

Peak relative displacement, zeta=0% (undamped), dt=0.003125s (finest): max|u(t)|, SDOF recurrence for piecewise-linear-interpolated forcing (undetermined coefficients), zeta=0% = 397.874575 mm

MDOF §2 — peak roof Ux |modal-rayleigh| relative diff %, dt=0.0125s (finest): |peakModal - peakRayleigh| / max(|peakModal|,|peakRayleigh|) * 100, both formulations calibrated to 3% at this model's own two dominant eigenfrequencies (mirrored from mdof.quantities below for §5/§6 rendering — see mdof block for the full MDOF model description) = 0 %

*Reference is NOT transcribed from any external table — it is the exact solution of the damped SDOF equation of motion (ü + 2*zeta*omega*u̇ + omega²*u = -üg(t)) for the SAME piecewise-linear-interpolated ground-motion samples the app's THA time series assumes (linear between samples), re-derived from scratch via undetermined coefficients (free-vibration term + forced response to a linear-in-tau interval force, zero IC) in benchmarks/dynamic/TH1_sdofAnchor.ts. De-risked (SH1 lesson, devplan §12.11, 'hand-derivation isn't trustworthy alone') two ways BEFORE any engine run: (1) an internal algebraic identity (d/dtau[unit-ramp response] == unit-step response, a basic LTI fact not assumed anywhere in the derivation) falls out exactly; (2) internal/TH1_closedform_selfcheck.ts cross-validates the closed form against an independent fixed-fine-substep RK4 integrator (completely different numerical method, shares no formula) to ~1e-11% before the engine runner was ever written. Because both STRIX and the closed form consume the identical discretized (piecewise-linear) excitation, any difference between them is pure Newmark time-integration error — which the average-acceleration method (unconditionally stable, 2nd-order accurate) must shrink as O(dt²) as dt -> 0.*

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Peak relative displacement, zeta=5%, dt=0.003125s (finest) (mm)	zeta=0.05, dt=0.003125	200.782798	200.786209	−0.0017%	PASS
Peak relative displacement, zeta=0% (undamped), dt=0.003125s (finest) (mm)	zeta=0, dt=0.003125	397.855333	397.874575	−0.0048%	PASS
MDOF §2 — peak roof Ux |modal-rayleigh| relative diff %, dt=0.0125s (finest) (%)	modal vs rayleigh, dt=0.0125 (finest)	1.88×10⁻⁴	0	+0.00019%	PASS

§7 §1 DT CONVERGENCE (VS CLOSED FORM) + §2 DT SELF-CONSISTENCY (MODAL VS RAYLEIGH)

CASE	PEAK DIFF (%)	DIFF
zeta=0, dt=0.1	43.187415	+43.1874%
zeta=0, dt=0.05	11.607091	+11.6071%
zeta=0, dt=0.025	2.940893	+2.9409%
zeta=0, dt=0.0125	0.737395	+0.7374%
zeta=0, dt=0.00625	0.184475	+0.1845%
zeta=0, dt=0.003125	0.046126	+0.0461%
zeta=0.05, dt=0.1	35.080211	+35.0802%
zeta=0.05, dt=0.05	9.001591	+9.0016%
zeta=0.05, dt=0.025	2.277168	+2.2772%
zeta=0.05, dt=0.0125	0.571346	+0.5713%
zeta=0.05, dt=0.00625	0.142864	+0.1429%
zeta=0.05, dt=0.003125	0.035719	+0.0357%
mdof dt=0.2	0.005259	+0.0053%
mdof dt=0.1	0.00937	+0.0094%
mdof dt=0.05	0.007882	+0.0079%
mdof dt=0.025	0.002645	+0.0026%
mdof dt=0.0125	1.8774×10^{-4}	+0.00019%

§1 (zeta=0/5%, first 12 rows): as dt halves six times (T/10 → T/320), the peak-displacement error against the independent closed form shrinks by almost exactly 4× each step — a measured convergence order of `2.000` at the finest dt pairs for both damping ratios, textbook confirmation of the average-acceleration Newmark method's theoretical $O(dt^2)$ accuracy. The finest-dt error itself is a few thousandths of a percent, orders of magnitude inside the 0.01% gate. **§2** (last 5 rows, "mdof dt=…"): because modal and Rayleigh damping are two genuinely different numerical formulations (not one exact answer vs an approximation), their gap does not vanish to zero — it shrinks with dt and floors out at a small residual from the few off-resonance modes (3–9) each scheme treats slightly differently. At the finest swept dt (0.0125s) the gap is `0.00019%`, about 2,600× inside the 0.5% self-consistency gate. (Note: the MDOF excitation amplitude a0 was reduced 10× from the value first used in devplan §12.14 purely for report-card readability — the model is linear-elastic, so this rescales the roof peak displacement proportionally without moving the gating diff% at all, which is a pure ratio of two amplitudes.)

§8 CONCLUSION

PASS

The app's production THA path on the STRIX engine reproduces an independently re-derived (and RK4-cross-checked) SDOF closed form to a few thousandths of a percent at the finest dt, with the theoretically expected O(dt²) convergence order measured directly across six dt halvings and two damping ratios (§1, **PASS**). On a real multi-mode structure, the production modal and Rayleigh damping formulations agree to 0.00019% at the finest dt — about 2,600× inside their self-consistency gate — confirming both damping paths, the period-based Rayleigh calibration, and the production node/element/mass emission functions are all internally consistent (§2, **PASS**). TH1 defends the dynamic time-integration layer of the nonlinear analysis pipeline; the element (SH1) and static nonlinear driver (SP1) layers are verified on their own cards.

§9 REFERENCES & ANALYSIS FILES

1. Newmark, N. M. (1959) — *A Method of Computation for Structural Dynamics*, ASCE Journal of the Engineering Mechanics Division 85(3) (average-acceleration method, γ=0.5, β=0.25, unconditionally stable, O(dt²) accurate — the closed-form comparison target for §1's convergence order).
2. Chopra, A. K. — *Dynamics of Structures* (SDOF piecewise-linear-interpolated-excitation recurrence form, re-derived independently in benchmarks/dynamic/TH1_sdofAnchor.ts, not transcribed from any table).
3. SM5 (devplan §12.1, refsolutions/SM5.json) — Bathe & Wilson ten-bay nine-storey frame, `buildBatheWilsonModel`, reused verbatim for §2 (already validated <0.001% against published eigenvalues).
4. STRIX production THA builder `buildThaTcl` run on the STRIX engine (THA route) — the full app script, including its gravity stage and recovery ladder; see benchmarks/dynamic/TH1_sdofAnchor.ts and TH1_mdofSelfConvergence.ts file headers for why those stages do not act on these linear problems.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · TH1/` `sha256 (pending publish)`